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Implementation of a Solid Target Production Facility

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Abstract. The desire to utilize long-lived PET isotopes in Australia has significantly increased over the years and several research projects for labelling of peptides, proteins and biomolecules, including labelling of recombinant antibodies has been restricted due to the limited availability of suitable isotopes. This need has led to the recent installation and commissioning of a new facility dedicated to fully automated solid target isotope production, including ^{124}I, ^{64}Cu, ^{89}Zr and ^{86}Y at the Austin Health Centre for PET.

Keywords: ^{124}I, ^{64}Cu, Solid Target, PET isotopes, IBA Nirta, Terimo, Pinctada, TEMA Sinergie.
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INTRODUCTION

Following the commissioning in 2008 of a new IBA 18/9 cyclotron along with a 2-meter external beam transport line, we were able, in 2011, to install a fully automated solid targetry facility for the production of ^{124}I, ^{64}Cu, ^{89}Zr and ^{86}Y. Our aim was the implementation of a safe and cost effective system utilizing commercial equipment from manufacturers who were able to demonstrate convincing data for the production of the intended radionuclides (Table 1).

TABLE 1. Solid Target Production of Selected Radionuclides

Isotope (t½)	Target	Production	Recovery
^{124}I (4.2 d)	Enriched oxide melted on disk	^{124}Te(p,n)^{124}I	Thermochromatographic separation
^{64}Cu (12.7 h)	Enriched metal electrodeposition	^{64}Ni(p,n)^{64}Cu	Acid dissolution, ion exchange
^{89}Zr (3.3 d)	Disposable foil or electrodeposition	^{89}Y(p,n)^{89}Zr	Acid dissolution, ion exchange
^{86}Y (14.7 h)	Enriched oxide compressed on disk	^{86}Sr(p,n)^{86}Y	Acid dissolution, ion exchange

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Choosing the Right Equipment

Solid Target: Two commercial solid target systems were readily available to plug onto our IBA beam line. Although the NIRTA Solid Target system proposed by IBA was significantly dearer than the other, it appeared more versatile in its design allowing irradiation of simple 24 mm diameter disks. In addition, data were available in the literature for the production of all 4 proposed radionuclides using NIRTA solid target (2, 3, 4, 5).

Hotcells: Several hotcell manufacturers responded to our tender; however TEMA Sinergie (6) was the only one who proposed a customized design fitting precisely to our requirements, especially in terms of functionality for operating the enclosed processing modules and computer monitoring with cGMP compliance for Audit Trail & Alarm Log File.

Transfer System: No appropriate Solid Target Transfer System (STTS) for transferring the irradiated target disk from the NIRTA solid target to the dedicated TEMA hotcell was available on the market. Fortunately, after discussing our requirements with different manufacturers, TEMA Sinergie agreed to design and build such a system, including a docking station located underneath the NIRTA solid target, a system for transferring the irradiated disk from the solid target to the hotcell and a disk recovery system inside the hotcell.

Target Processing Modules: Several radiochemistry processing modules existed on the market, but again only IBA manufacturer provided strong data supporting the successful processing of ^{124}I (TERIMO module) (7) as well as evidence of the design of a versatile module (Pinctada Metal module) compatible with the dissolution/purification of a variety of metallic targets.

RESULTS

Solid Target: Although the installation of the NIRTA solid target was straight forward and took less than a day, optimization of the beam shape with homogeneous distribution onto the target disk required significantly more work, including the physical re-alignment of the 2 meter beam line (Figure 1).

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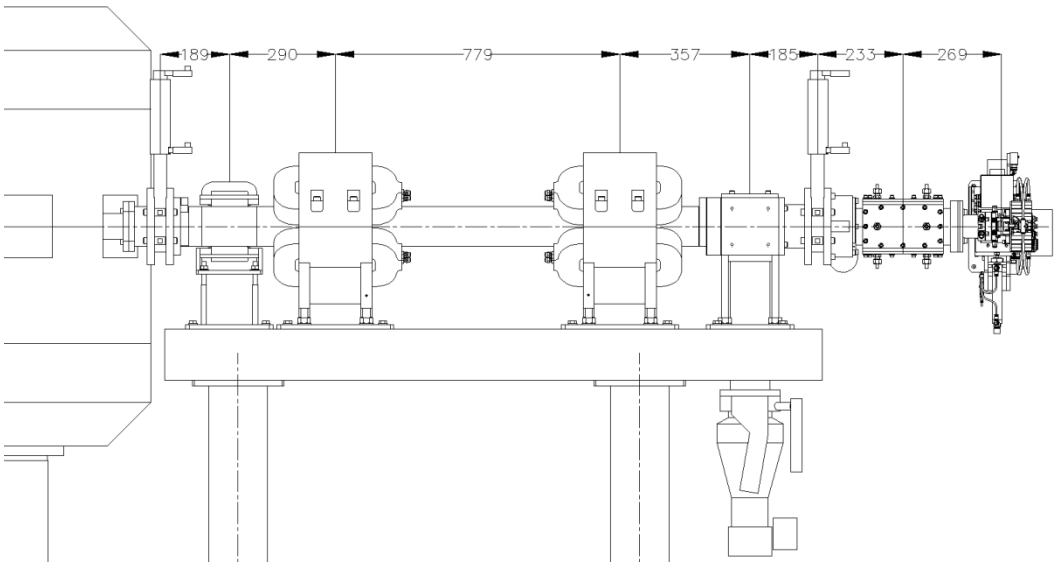


FIGURE 1. Schematic of IBA beamline.

The beamline consists of two steering dipole magnets for changing the beams position in the X and Y axis and a quadrupole doublet for beam focusing. Monitoring of the beam profile is achieved via a four-finger collimator and calibration of the quadrupole settings was achieved by a comparison of collimator readings with beam profiles burnt into acrylic discs loaded into the NIRTA solid target irradiated for 10 seconds at 1μA.

The IBA NIRTA Solid Target allows the loading of up to 3 target disks into a charger for sequential irradiation without re-entering the vault (Figure 2).



FIGURE 2. NIRTA solid target.

Eight I-124 productions, ranging from 3 to 18 hours and 10 to 15 μA as well as six Cu-64 productions ranging from 2 to 6 hours and 20 to 35μA have been performed without problem. Both the beam line parameters and the target current are fully

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controlled from the cyclotron computer and were very stable, even after 18 hours of irradiation.

Hotcells: Three TEMA hotcells were installed side by side in a room adjacent to the solid target bunker with an 800mm technical corridor at the back (Figure 3). The larger hotcell dedicated to ^{124}I is shielded with 100mm of lead and equipped with two telemanipulators as well as a shielded drawer for insertion/extraction of equipment or lead pots. The adjacent 2 hotcells dedicated to ^{64}Cu, ^{89}Zr and ^{86}Y are 75mm thick and equipped with one manual teletong and a shielded drawer for insertion/extraction of small items as well as a vial-drop with independent lead pot extractor. All hotcells feature a dose calibrator, a touch screen display for control and sensor monitoring and a ventilation system with HEPA filter, providing air quality to Class II and GMP Class C environment.

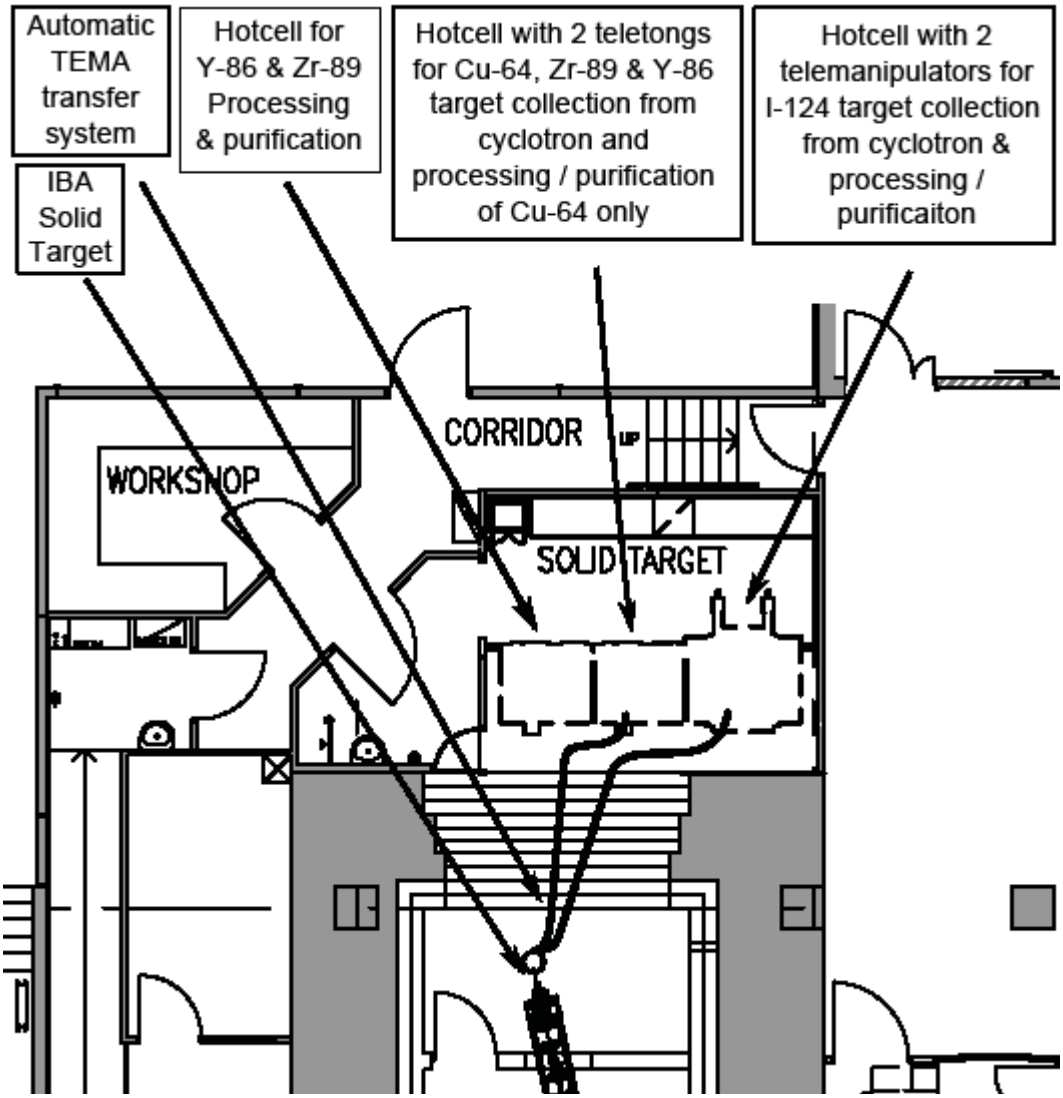


FIGURE 3. Solid Targetry Facility layout showing Hotcells, Target Transfer System and Solid Target.

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Transfer System: The TEMA STTS is operated from a touch screen display conveniently located on the front of one of the hotcells and features safety software interlocks between the IBA irradiating station and TEMA processing hotcell. A rotating table equipped with six shuttles is installed underneath the NIRTA target, ready to receive sequentially irradiated target disk. Each shuttle, driven by a fully automated pneumatic system, can be closed with a lid and transferred up to the selected hotcell. Inside the hotcells, the telemanipulator or teletong is used to open the shuttle lid and recover the irradiated disk.

Target Processing modules: The system consist of 2 modules, one dedicated to the production of I-124 (TERIMO) and one dedicated to the production of other metallic targets such as Cu-64 (Pinctada). The latter also includes an electroplating system for electro-deposition of Ni-64 onto a support disk.

The TERIMO module is a closed system with an integrated charcoal filter. The system allows for the preparation of the target disk by melting $^{124}TeO_2$ powder into the target disk recess. After irradiation the same TERIMO module is used to recover ^{124}I by distillation at a furnace temperature of 780°C and collection into 1 mL of alkaline solution.

The IBA Pinctada Metal module is a closed system for the dissolution and purification of metallic target. The system allows for the clamping of the target disk for hot acidic dissolution and sequential ion exchange separation/purification of the radionuclides.

CONCLUSIONS

A commercially available, fully automated, solid target facility was successfully installed at Austin Health. The functionality of all equipment has met the given specifications and batch productions of ^{124}I & ^{64}Cu have shown results in good accordance with theoretical calculations.

ACKNOWLEDGMENTS

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